

# Real-Time DC-Link Capacitor Stress Reduction in Two-Level Three-Phase Inverters Using Adaptive Multi-Carrier PWM Method

Track : 4 Motor Drive and Control

**Abstract**—This paper presents a novel multi-carrier pulse width modulation (PWM) technique for two-level three-phase inverters to reduce the current stress of the DC-link capacitors, thereby improving system reliability and extending capacitor lifetime. Unlike conventional interleaving approaches used in parallel inverter systems or pulse synchronization in multi-stage converters, the proposed method applies adaptive interleaving between the inverter legs of a single two-level three-phase inverter unit. The carrier phase shifts are dynamically updated in each control cycle based on real-time duty cycles and leg currents, enabling effective redistribution of switching events and minimizing capacitor RMS current. Analytical and experimental results confirm the effectiveness of the proposed PWM strategy in significantly reducing ripple current up to 27% and enhancing the reliability and lifetime of the DC-link capacitors.

**Index Terms**—Motor drive, DC-link current ripple, sinusoidal pulse width modulation, multi-carrier PWM, voltage source inverter.

## I. INTRODUCTION

Two-level three-phase voltage source inverters (VSIs) have still remained as one of the dominant topologies in industrial motor drives thanks to their reduced component count, simplicity, ease of control, and cost-effectiveness [1], [2]. Despite these advantages, reliability concerns continue to pose a significant challenge. One of the main reliability concerns is caused by DC-link capacitors, reportedly responsible for approximately 30% of industrial power converter failures [3]. A key contributor to these failures is the ripple current flowing through the capacitor, which induces thermal stress and accelerates aging, thereby reducing its operational lifetime [4], [5]. Accordingly, various techniques have been investigated to reduce the ripple current magnitude and spectral content to enhance the reliability and enhance the lifetime of the DC-link capacitors [6]–[20].

Among the various ripple reduction strategies, interleaving in multi-inverter systems has demonstrated promising results [8]–[12]. Several studies show that optimal phase shifting and current ripple synchronization among parallel inverters or multiphase machines effectively reduce DC-link ripple and enhance capacitor longevity. Similarly, in two-stage inverter systems, ripple mitigation has been achieved through dynamic

coordination of switching sequences between the front-end and inverter stages using interleaved carriers [14], [15].

In contrast to these approaches, which rely on multiple inverter legs or cascaded stages, this paper proposes a novel modulation strategy that applies adaptive interleaving between legs of a single two-level three-phase VSI. In the proposed method, leg-specific carrier phase shifts are dynamically updated in each control cycle based on the instantaneous duty ratios and phase currents. This real-time adaptation redistributes the switching instants to minimize the total RMS current stress on the DC-link capacitor, thereby enhancing its lifetime. Notably, this strategy requires no additional hardware, such as parallel inverters or a front-end converter, making it a practical and low-complexity solution for ripple reduction. While the concept builds on interleaving principles, implementing it within a single inverter introduces new challenges due to the time-varying nature of phase currents and the interaction of three duty ratios and three leg currents. That is to say, determining the optimal carrier phase shifts in real time is computationally intensive due to the high dimensionality of the problem. Therefore, the paper introduces a look-up table (LUT) that is constructed offline to estimate the carrier phase shifts regarding the duty ratios and leg currents covering all the possible operating points of the inverter. The comparison of the proposed method with existing literature is summarized in Table I. Unlike prior approaches that rely on parallel inverters, multiphase drives, or additional DC–DC stages, the proposed method achieves effective ripple reduction within a single two-level three-phase VSI. It requires no extra hardware, with the only limitation being the need for an offline look-up table (LUT) to enable adaptive carrier phase shifting in real time.

The remainder of this paper is organized as follows. Section II analyzes the capacitor current behavior in a two-level three-phase voltage source inverter and provides a mathematical formulation for RMS stress evaluation. The conventional single-carrier PWM method is reviewed in Section III, while the proposed multi-carrier PWM technique is detailed in Section IV. Section V presents the comparison of the conventional method and proposed method. Section VI discusses implementation aspects, including lookup table construction. Experimental validation of the technique is provided in Section VII. Finally, the conclusion is drawn in Section VIII.

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TABLE I  
DC-LINK CAPACITOR RIPPLE REDUCTION STRATEGIES IN LITERATURE

| Ref.      | Topology                      | Approach                         | Evaluation                                                       |
|-----------|-------------------------------|----------------------------------|------------------------------------------------------------------|
| [11]      | Integrated modular drive      | Multi-phase interleaving         | Requires multi-phase IMM; Applying constant carrier phase shifts |
| [12]      | Two parallel 2L VSIs          | Hybrid carrier-based PWM         | Needs two inverters; Updating reference signals                  |
| [14]      | Single-phase 2-stage inverter | Carrier shaping in DC-DC stage   | Needs pre-stage converter; Altering carrier signals              |
| [15]      | Three-phase 2-stage inverter  | Generalized carrier-shift method | Needs pre-stage converter; Altering carrier signals              |
| This work | Single 2L-3Ph VSI             | Real-time multi-carrier PWM      | No extra hardware; Applying adaptive carrier phase shifts        |

## II. DC-LINK CURRENT ANALYSIS OF THE INVERTER

Fig. 1 illustrates a conventional two-level three-phase (2L-3Ph) voltage source inverter (VSI) along with its simplified equivalent circuit.

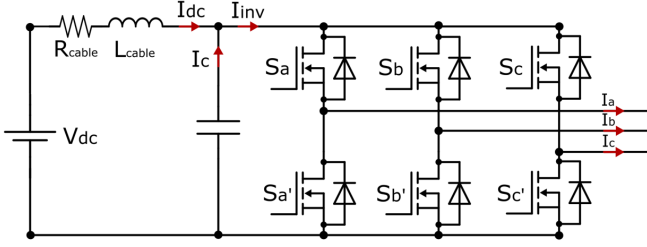


Fig. 1. Two-level three-phase voltage source inverter.

The equivalent model highlights the DC-link current path, under the assumption that all high-frequency components generated by inverter switching are fully absorbed by the DC-link capacitor. Therefore, mathematically, the instantaneous capacitor current  $[i_{cap}(t)]$  can be expressed as the sum of individual phase currents  $[i_x(t) \text{ for } x \in \{a, b, c\}]$  multiplied by their respective switching functions  $[S_x(t) \text{ for } x \in \{a, b, c\}]$ , minus the average DC component  $[\bar{i}_{dc}]$  as given by

$$i_{cap}(t) = \sum_{x \in \{a, b, c\}} i_x(t) S_x(t) - \bar{i}_{dc}. \quad (1)$$

The switching function for each phase is determined by comparing the phase reference voltage as

$$v_{x,ref}(t) = m_a \cos(\theta(t) - 2\pi(x-1)/3), \quad (2)$$

with the triangular carrier waveform as

$$\text{tri}(2\pi f_{sw}t - \theta_{x,carrier}), \quad (3)$$

and can be calculated as

$$S_x^{conv}(t) = \begin{cases} 1 & \text{if } v_{x,ref}(t) > \text{tri}(2\pi f_{sw}t - \theta_{x,carrier}), \\ 0 & \text{otherwise,} \end{cases} \quad (4)$$

where  $m_a$  is the modulation index,  $f_{sw}$  is the switching frequency,  $\theta_{x,carrier}$  is the carrier phase of legs for  $x \in \{a, b, c\}$ .

In this paper, a hierarchical approach, covering both switching-period and fundamental-period dynamics of inverter operation, is proposed to analyse the DC-link current. First, at the switching period ( $T_{sw} = 1/f_{sw}$ ), the instantaneous RMS current is computed as:

$$I_{cap,rms}[k] = \sqrt{\frac{1}{T_{sw}} \int_{(k-1)T_{sw}}^{kT_{sw}} i_{cap}^2(t) dt} \quad \text{for } k = 1, 2, \dots, N_{sw} \quad (5)$$

where  $N_{sw} = f_{sw}/f_1$  is the number of switching intervals within one fundamental period. During this period, both the reference voltages and phase currents are assumed to vary slowly as the switching frequency is much faster than the fundamental frequency, as given:

$$v_{x,ref}[k] \approx m_a \cos\left(\frac{2\pi k}{N_{sw}} + \theta_{fund} - \frac{2\pi(x-1)}{3}\right) \\ i_x[k] \approx I_{pp} \cos\left(\frac{2\pi k}{N_{sw}} + \theta_{fund} - \frac{2\pi(x-1)}{3} - \theta_{phase-angle}\right) \quad (6)$$

which allows the capacitor current waveform to be evaluated at quasi-static conditions within each switching cycle. Then, the steady-state fundamental-period RMS current can be calculated by combining all switching-period RMS, as given by:

$$I_{cap,rms}^{total} = \sqrt{\frac{1}{N_{sw}} \sum_{k=1}^{N_{sw}} I_{cap,rms}^2[k]} \quad (7)$$

This formulation shows that minimizing each  $I_{c,rms}[k]$  directly reduces  $I_{c,rms}^{total}$ , which means that individual minimization during each switching interval leads to fundamental-RMS minimization.

## III. CONVENTIONAL SINGLE-CARRIER-BASED PWM

The conventional single-carrier-based PWM scheme employs identical triangular carrier signals for all three phases, given in Fig. 2. The conventional method's performance is fundamentally constrained by the uncontrolled switching events, which prevent the optimum harmonic cancellation between phases regarding each switching interval. An example operation, the key waveforms are given as shown in Fig. 3.

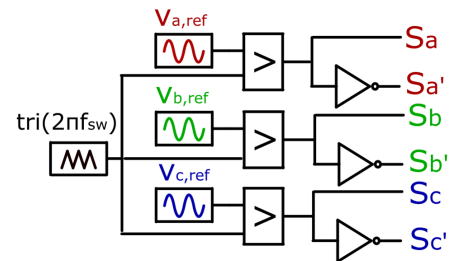


Fig. 2. Conventional Single-Carrier PWM

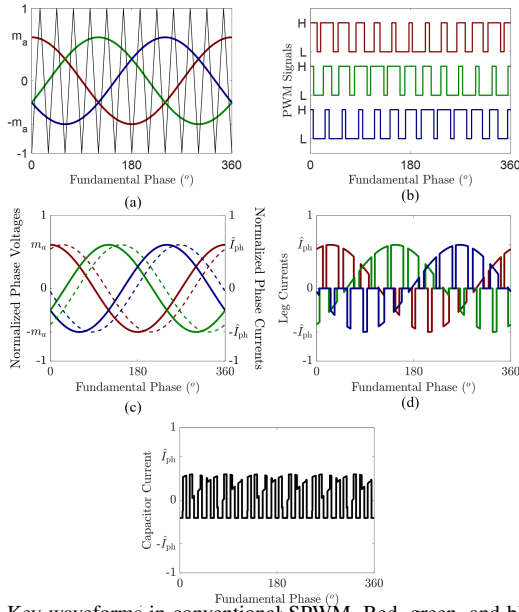


Fig. 3. Key waveforms in conventional SPWM. Red, green, and blue curves correspond to phases A, B, and C, respectively. (a) Reference signals and common triangular carrier waveform. (b) Resulting PWM gate signals for each inverter leg. (c) Normalized phase voltages (solid) and phase currents (dashed). (d) Leg currents. (e) DC-link capacitor current waveform.

#### IV. PROPOSED MULTI-CARRIER PWM TECHNIQUE

The proposed method introduces an adaptive interleaving by applying phase-shifted triangular carrier signals, which is updated within each switching period, to each leg of a two-level three-phase inverter, as the proposed PWM scheme is shown in Fig. 4.

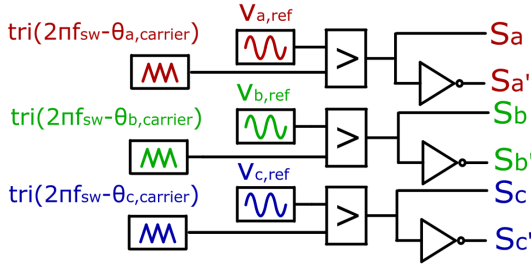


Fig. 4. Proposed Multi-Carrier PWM

Each phase is assigned an individual carrier waveform with a phase displacement  $\delta_x$  ( $x \in \{a, b, c\}$ ), where  $\delta_a = 0$ ,  $\delta_b = \theta_{b,carrier}$ , and  $\delta_c = \theta_{c,carrier}$ . An example operation, the key waveforms are given as shown in Fig. 5.

The goal is to dynamically determine the optimal phase shifts  $\theta_{b,carrier}^{opt}$  and  $\theta_{c,carrier}^{opt}$  in each switching cycle to minimize the RMS current through the DC-link capacitor. The optimal phase shifts ( $\theta_{b,carrier}^{opt}, \theta_{c,carrier}^{opt}$ ) are obtained by minimizing the capacitor RMS current over the switching period as

$$(\theta_{b,carrier}^{opt}[k], \theta_{c,carrier}^{opt}[k]) = \arg \min_{\theta_b, \theta_c} \left( \int_{(k-1)T_{sw}}^{kT_{sw}} i_{cap}^2(t) dt \right) \quad (8)$$

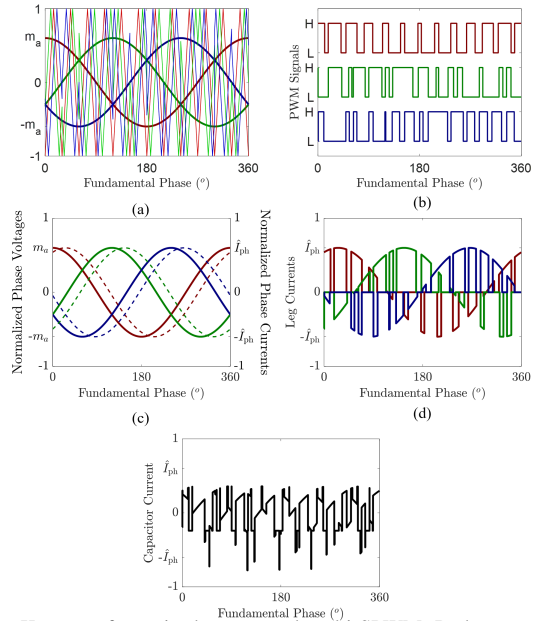


Fig. 5. Key waveforms in the proposed multi SPWM. Red, green, and blue curves correspond to phases A, B, and C, respectively. (a) Reference signals and common triangular carrier waveform. (b) Resulting PWM gate signals for each inverter leg. (c) Normalized phase voltages (solid) and phase currents (dashed). (d) Leg currents. (e) DC-link capacitor current waveform.

The optimization is done offline with respect to distinct duty ratios and phase current values, then it is implemented in the controller, which will be discussed in the following sections.

#### V. PERFORMANCE ASSESSMENT OF THE PROPOSED MULTICARRIER PWM METHOD

In this section, the effectiveness of the proposed multicarrier PWM method is evaluated in terms of reducing the RMS current stress on the DC-link capacitor. The assessment is conducted through numerical calculations. Two distinct operating points are selected for both the conventional and the proposed systems, and the corresponding RMS values of the capacitor current at the switching frequency are calculated. The comparison results are presented in Fig. 6.

Then, Fig. 7 illustrates the root mean square (RMS) value of the DC-link capacitor current as a function of the modulation index ( $m_a$ ) and power factor ( $p.f.$ ), comparing the conventional single-carrier PWM with the proposed multicarrier modulation strategy. These results indicate that the proposed method achieves a notable reduction in RMS current (up to %27) across a wide range of operating conditions. This reduction leads to decreased thermal stress on the capacitor, which in turn contributes to an extended capacitor lifetime.

The remaining sections will be provided in the full paper.

#### VI. IMPLEMENTATION METHODOLOGY

#### VII. EXPERIMENTAL VALIDATION

To validate the effectiveness of the proposed multi-carrier PWM technique, a laboratory-scale experimental setup was established using a two-level three-phase inverter, connected to an induction machine, as given in Fig. 8.

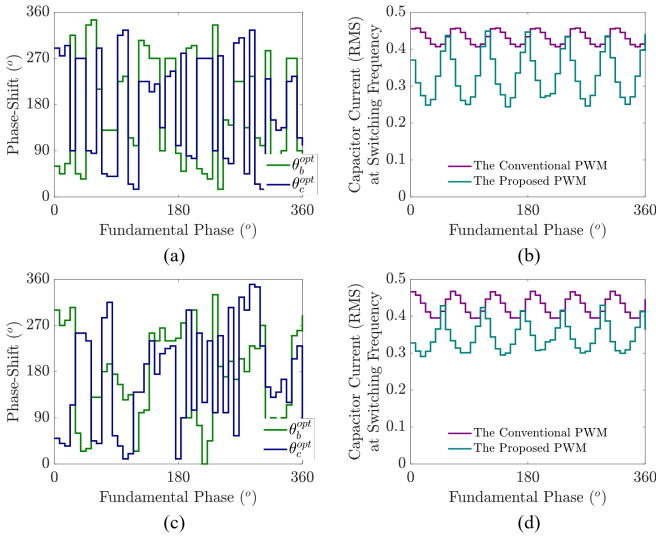


Fig. 6. Capacitor current analysis under various modulation conditions. (a) Optimum carrier phase shift for  $m_a = 0.6$ ,  $I_{ph} = 1$  p.u., and power factor  $p.f = 0.9$ . (b) RMS capacitor current at the switching frequency for the same condition. (c) Optimum carrier phase shift for  $m_a = 0.8$ ,  $I_{ph} = 1$  p.u., and power factor  $p.f = 0.95$ . (d) RMS capacitor current at the switching frequency comparison for the same condition.

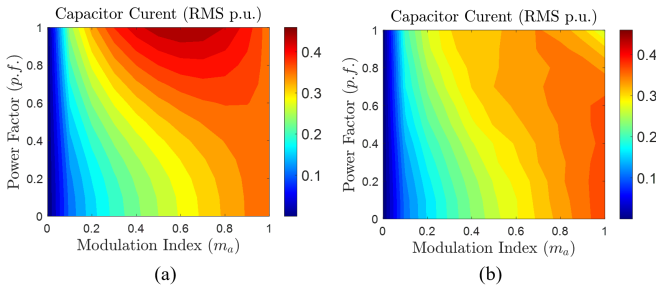


Fig. 7. RMS capacitor current across varying  $m_a$  and power factor values. a) The conventional single-carrier PWM. b) The proposed multicarrier PWM method

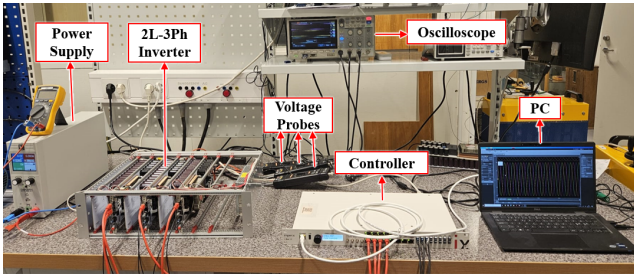


Fig. 8. Experimental setup of 2-level 3-phase inverter system.

## VIII. CONCLUSION AND FUTURE WORK

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